Higher National Diploma in Engineering (Electrical)
Fourth Year, First Semester Examination – 2021
EE 3204– Electrical Energy Management

Instructions for Candidates:
Answer **Five (05)** questions.
Each question carries equal marks

No. of questions:06
No. of pages : 04
Time : **03 Hours**

Q1.

(a) State advantages and disadvantages of Auto Transformers over Double-wound Transformers?

(03 Marks)

(b) What are the main types of three phase winding connections in three phase transformers? Give their special characteristics and applications.

(04 Marks)

(c) Briefly explain the available vector grouping in 3-ϕtransformers. Give reasons for above vector grouping.

(06 Marks)

(d) A 120 kVA transformer has its maximum efficiency of 0.985 at full load unity power factor. During the day it is loaded as follows:

8.00 a.m. to 1.00 p.m	-	75 kW, 30 kVar
1.00 p.m. to 6.00 p.m	-	85 kW, 45 kVar
6.00 p.m. to 1.00 a.m	-	35 kW, 25 kVar
1.00 a.m. to 8.00 a.m	-	10 kW, 05 kVar

Determine the all-day efficiency of the transformer.

(07 Marks)

[Total 20 Marks]

Q2.

(a) Explain the principle of good Lighting design?

(03marks)

(b) What are the factors are to be consider while designing a good Lighting Syatem?

(04Marks)

(c) Define the term Luminance? Explain Utilization factor (U.F) with the equation?

(06 Marks)

(d) List out the types of Lighting Lamps? Explain the construction & working principle of Low-pressure Mercury Fluorescent Tube?

(07 Marks)

[Total 20 Marks]

Q3.

(a)

(i) What are the two general types of polyphase induction motors?

(ii) State the differences between them.

(iii) List down the advantages and disadvantages of the types mentioned in above Question?

(04 Marks)

(b) What are the effects on performance of poly phase induction motor while the supply voltage is unbalanced?

(03 Marks)

(c)

(i) List down three different speed control mechanism of poly phase induction motor.

(ii) State three starting methods of poly phase induction motors.

(05Marks)

(d) A 3-phase 15 HP(11.0325 kW), 460 V, four pole, 60 Hz, 1728 rpm induction motor delivers full output power to a load connected to its shaft. The windage and friction loss of the motor is 750 W.

Determine the,

(i) Mechanical power developed

(ii) Slip of the motor

(iii) Air gap power
(iv) Rotor copper loss

(08 Marks)

[Total 20 Marks]

Q4.

(a) What is electric tariff? Discuss the objectives of Tariff?

(06 Marks)

(b) Write short notes on the following:
   i. Two-part tariff
   ii. Power factor tariff
   iii. Three-part tariff

(06 Marks)

(c) An industry has a maximum load of 240 KW at 0.8 p.f .lagging with an annual consumption of 50,000 units. The tariff is Rs. 70 per kVA of maximum demand plus 20 cents per unit. Calculate the flat rate of energy consumption. What will be the annual saving, if the power factor is raised to unity?

(08 Marks)

[Total 20 Marks]

Q5.

(a) What is Lagging, Leading and Unity Power factor? Explain

(06Marks)

(b) What are true power, apparent power and reactive power? How to maintain the power factor at a higher value?

(06 Marks)

(c) Discuss the various methods for power factor improvement?

(08 Marks)

[Total 20 Marks]

Q6.

(a) What do you mean by energy audit? Explain the role of energy auditor and the things to be considered while auditing?

(08 Marks)

(b) Explain the elements of energy management in detail? What are the benefits of energy management?

(06 Marks)

(c) Write a short note on Demand side Management & Supply Side Management?

(06 Marks)

[Total 20 Marks]